SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

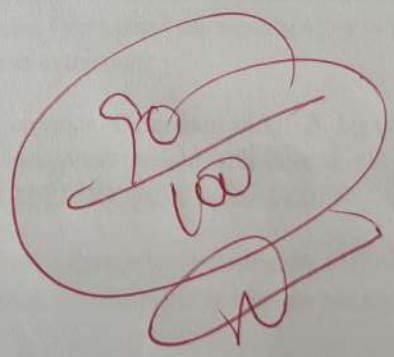
Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I M.E.Koothik(192521175) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 20	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 25	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

90

90
100

Scenario: INTERCEPTED SUSPICIOUS MESSAGE

Given ciphertext,

"GSRH RH Z HVXIVG.

The structure of the ciphertext suggests a simple substitution cipher. The most likely cipher is the Atbash cipher

a) Deduce the most likely original message

Answer:

The original Message is: "This is A SECRET"

The cipher is Atbash, a types of monoalphabtic substitution cipher.

In Atbash, each alphabet character is replaced by its corresponding character from the ~~the~~ reverse alphabet

Abtash substitution Table:

Plaintext	A	B	C	D	E	F	G	H	I
ciphertext	Z	Y	X	W	V	U	T	S	R
Plaintext	J	K	L	M	N	O	P	Q	R
Ciphertext	Q	P	O	N	M	L	K	J	I
Plaintext	S	T	U	V	W	X	Y	Z	
Ciphertext	H	G	F	E	D	C	B	A	

character by character decoding

G S R H R H Z H V X I V G
↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
T H I S I S A S E C R E T

Hence, GSRH · RH ZHVXIVG → THIS IS A SECRET

b. Justification

The answer can be justified using word patterns, letter substitution, and known cipher characteristics

1) The text has 4 words

GSRH RH Z HVXIVG
4 2 1 6

The proposed plain text has exactly the same structure
THIS IS A SECRET
4 2 1 6

substitution ciphers preserve the no of characters and word boundaries.

2) one letter word
The ciphertext contains 'Z'
common one letter english words are A and I
Under Atbash Z → A.
This provides strong evidence for the plaintext word A.

3. Consistevnt Substitution

In a substitution cipher, one ciphertext character consistently represents one plain character

GSRH → THIS

4. Reverse alphabet uses the reverse alphabet All the maring in the ciphertext follow the revese alphabet reationship.

Cipher characteristics.

feature	Atbash cipher
Type	substitution cipher
Category	monoalphabet
Key Mechanism	reverse alphabet
Word Spaces	Preserved
letter positions	Preserved

Security observation: Atbash is not suitable for protecting sensitive modern communication because it An uses a fixed substitution pattern An attacker who identifies the cipher can easily decrypt the message

Conclusion:

The intercepted ciphertext GSRH RH Z HXXIVG is an Atbash cipher. The decoded message is "THIS IS A SECRET"

The deduction is supported by word length one-letter word analysis, consistent substitution reverse alphabet characteristics.

Scenario: SUSPICIOUS DIGIT ASSET INVESTIGATION

A company logo appears normal, but investigators suspect that confidential information has been embedded inside it this is possible case of image steganography.

a. How can investigation detect hidden data?

1. Understand the suspected technique steganography is the process of hiding secret information inside an apparently normal file.

an image:

Cover Image + Secret Data

↓

steganographic embedding

↓

Stego Image.

The stego image should appear normal to an ordinary viewer for example information can be hidden by modifying the least significant bits (LSBs) of image pixes

2) Preserve the evidence

Before investigation investigators should create a forensic copy of the image pixels

Suspicious image
↓
preserve orginal
↓
Create forensic copy
↓
Perform Analysis

The original file should not be unnessarily modified A cryptographic hash such as SHA-256 can be calculated.

Image
↓
SHA-256
Unique Hash Value

normal to our ordinary
hidden by modifying
pixels

This helps verify the integrity of the evide

3. File & metadata Analysis

* file formate &size
* Image Dfensions & color depth
* Compression Details
* creation / modifiation dates
* Metadata & embeddeal comments
* software Information.

Tool : exif Tool

4. Visual Inspection

* unusuae colors / noise
* pixel distortions
* strange borders / regions
* unexpected visual Differenee.

Investigation Workflow

Suspicious Image

↓

Preserve evidence

↓

Calculate SHA 256

↓

Metedata Analysis

↓

Visual Inspection

↓

LSB + statistical Analysis

File – structure Analysis

↓

Document findings

b. two techniques to strengthen Hidden communication

1. encrypt Before embedding

The secret message should be first encrypted using a Stong algorithm such as AES.

Secret message

↓

AES encryption

↓

Ciphertext

↓

Steganographic embedding

↓

Image

even if an attacker extracts the hidden data, it will remain unreadable without the encryption key.

2.) Use robust low-Distortion steganograph

Use a steganographic strategy that distributes hidden data carefully & minimizes changes to the image.

Secret Data

↓

encryption

↓

Data Distribution

↓

low-Distortion embedding

↓

stego Image.

Conclusion:

Hidden data in a company logo can be detected using mete data analysis, Visual inspection, LSB analysis, Statistical steg analysis and File-structure analysis Tools. Such as exif Tool, Binwalk, steg and steg-spoke can assist the.

Investigation: For Stranger Protection: encrypt the secret before embedding It and use robost low-distortion steganography.